# Nam Nguyen

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## Education

**California Polytechnic State University, San Luis Obispo, CA** January 2023 – June 2024
M.S. Computer Science, *With Distinction*
Advised by Prof. Jonathan Ventura

**California Polytechnic State University, San Luis Obispo, CA** September 2019 – December 2022
B.S. Computer Science, *Summa Cum Laude*

## Experience

**Deepnight Inc**, Research Scientist December 2025 – Present
- Developed a novel HDR tone-mapping technique for extremely low-light AI image denoising, improving dynamic range preservation, local contrast, and visual quality in extreme exposure conditions.
- Designed a novel synthetic data generation strategy to improve AI model robustness in challenging HDR scenarios.
- Developed a cross-platform quantization framework for efficient AI inference across NVIDIA and Qualcomm platforms.

**Remix Inc**, Computer Vision Lead July 2024 – November 2025
- Led the development of real-time neural rendering system converting multi-camera livestreams into immersive 3D environments for WebXR, optimizing reconstruction, rendering, and real-time inference pipelines for interactive VR experiences.
- Designed and implemented TeleForm, a model that lifts single panoramic frames into a 3D environment, enabling real-time 3D scene rendering at up to 120 FPS on client browsers.

**Cal Poly, San Luis Obispo**, Research Scientist July 2024 – June 2025
- Developed novel view synthesis methods for 3D reconstruction from sparse and unconstrained inputs.
- Explored efficient neural scene representations for real-time and photorealistic 3D rendering.
- Advised undergraduate and graduate students on 3D vision research.

**Remix Inc**, Computer Vision Engineer Intern September 2023 – March 2024
- Worked on novel view synthesis methods for real-time 3D reconstruction from a live panoramic video stream.

**Cal Poly, San Luis Obispo**, Graduate Research Assistant September 2021 – June 2024
*3D Computer Vision Lab*
- Worked on extending 3D view synthesis methods to improve performance with casually captured data, including a set of unconstrained images captured by smartphones, or a single image.

*Deep Learning GIS Lab*
- Worked on integrating deep learning in different applications of remote sensing.

## Service

Reviewer for WACV 2024, CVPR 2024, ECCV 2024
Guest Lecturer for CPE/EE 428 – Computer Vision Winter 2024

## Skills

Python, PyTorch, Tensorflow, AIMET, QNN, OpenCV, OpenGL, Three.js, ArcGIS, QGIS, GDAL

## Publications

1. Nam Nguyen. Instant HDR-NeRF: Fast Learning of High Dynamic Range View Synthesis With Unknown Exposure Settings. Master's Thesis, 2024.

2. Krti Tallam, Nam Nguyen, Jonathan Ventura, Andrew G. Fricker, Sadie Calhoun, Jennifer O'Leary, Mauriça

Fitzgibbons, Ian C. Robbins, and Ryan K. Walter. Application of deep learning for classification of intertidal eelgrass from drone-acquired imagery. *Remote Sensing*, 15(9):2321, 2023.

3. Lillian Eagan, Nam Nguyen, Angela Victoria Chen, Theresa Zhu, Seth Maxwell Johnson, Pranav Reddy Dumpa, Benjamin Jonson Geil, Jonathan Ventura, and Stefanie Zollmann. VR Panorama Inpainting: A Feasibility Study of High-Resolution View Synthesis for Virtual Reality. *2026 IEEE International Conference on Artificial Intelligence and eXtended and Virtual Reality (AIxVR)*, 2026.

## Presentations

1. Nam Nguyen, Edward Du, and Jonathan Ventura. HDR-NeRF--: Learning High Dynamic Range View Synthesis With Unknown Exposure Settings. Paper. XRNeRF: Advances in NeRF for the Metaverse Workshop, Conference on Computer Vision and Pattern Recognition (*CVPR*), Vancouver, Canada, 2023.

2. Shivam Ajisa, Edward Du, Nam Nguyen, Stefanie Zollmann and Jonathan Ventura. 3D Pano Inpainting: Building a VR Environment from a Single Input Panorama. Poster. The 31st IEEE Conference on Virtual Reality and 3D User Interfaces (*IEEE VR*), Orlando, FL, 2024.

3. Nam Nguyen, Jessica Baiza, Griffen Guizan, Camille Pawlak, Sara Arnold, Ryley Chase, Lexxie Crocker, Jenn Yost, Matt Ritter, Geoffrey A Fricker, and Jonathan Ventura. OpenCanopy: Leveraging aerial imagery and deep learning to delineate California's urban tree canopy. Poster. Southern California Data Science Day, *The 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, Long Beach, CA, 2023.